

PyGameMaker — Tutorial 1: First Steps

How to Use This Handout

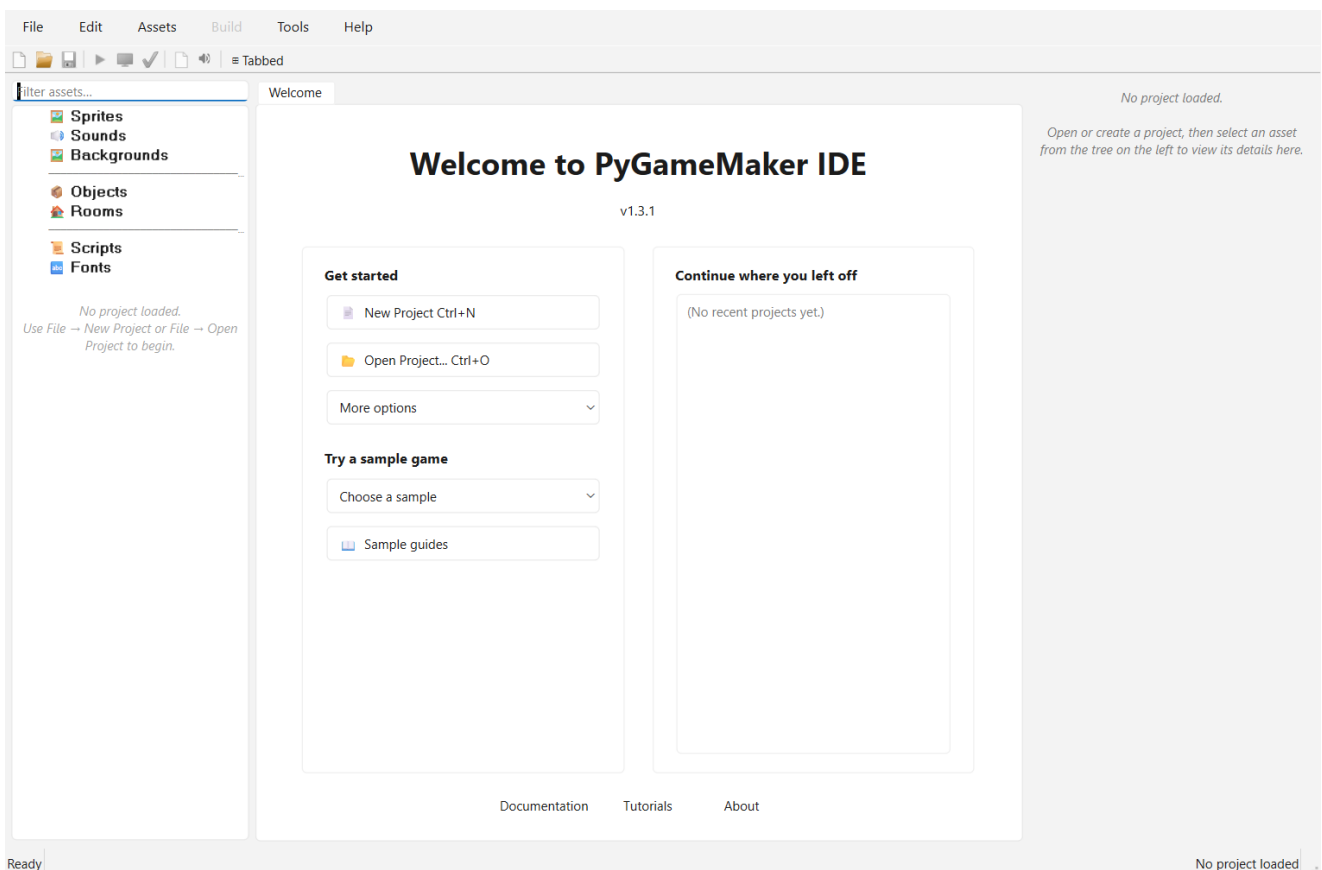
This handout follows the 4 pages of the in-app tutorial, found under **Help > Tutorials > First Steps**. Read each part here, then do the matching part on your screen. Tick the checkboxes as you complete each task, and use the **My Notes** section at the end to write down anything you want to remember.

Part 1: Welcome to PyGameMaker

PyGameMaker is a visual game-making program. You build 2D games by creating **assets** — sprites (images), objects (things that are programmed to behave a certain way), and rooms (levels) — and connecting them together. By the end of this tutorial, you will be able to:

- Navigate the PyGameMaker interface confidently
- Create and manage game assets (sprites, objects, rooms)
- Add behaviors to game objects using events and actions
- Test and export your own games

You can close the tutorial panel at any time with the "Close" button at the top. To open it again later, use **Help > Tutorials** from the menu.



- **Title Bar**
- **Close**
- **Fullscreen**
- **Minimize**

- **Menu Bar**
- **Toolbar**
- **Assets**
- **Editor Area**
- **Properties**

Part 2: The PyGameMaker Interface

Before building anything, take a moment to look around. The PyGameMaker window is divided into three main areas.

1. Asset List (left panel)

This is where every asset in your project lives, organized by type. Right-click any category here to create a new asset of that type.

- **Sprites** — the images and animations your game uses.
- **Sounds** — sound effects and music.
- **Objects** — the game entities that actually do things.
- **Rooms** — the levels or screens of your game.

2. Editor Area (center)

This is where you actually edit things. Double-click an asset in the tree, and its editor opens here in a tab.

- The **Sprite Editor** lets you view and draw sprites.
- The **Object Editor** lets you define what an object does, using events.
- The **Room Editor** lets you design a level by placing objects in it.

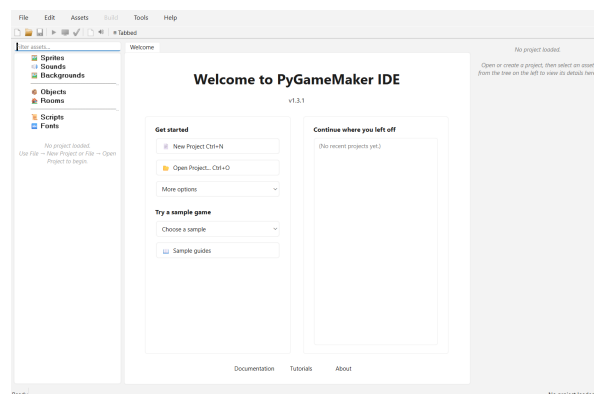
3. Properties Panel (right panel)

Shows the settings of whatever you currently have selected. When you are editing a room or an object, this is where some of its properties are displayed and where you change them.

Key Menus

- **File** — create, open, and save projects
- **Assets** — create and import game assets
- **Build** — test and export your game
- **Help** — tutorials and documentation

Press **F5**, or click the green triangle ► in the toolbar, at any time to quickly test your game!

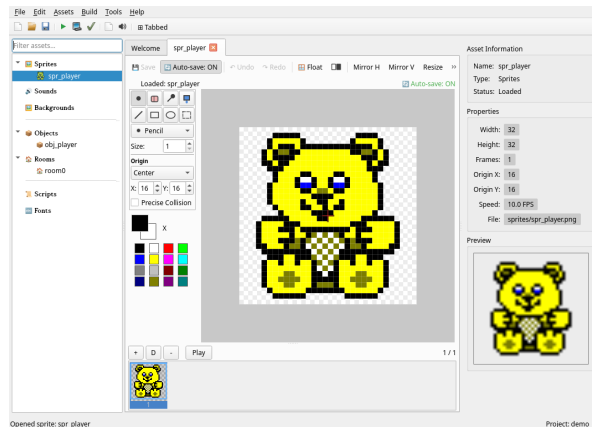


In the screenshot: **Assets** (left), **Editor Area** (center), **Properties** (right).

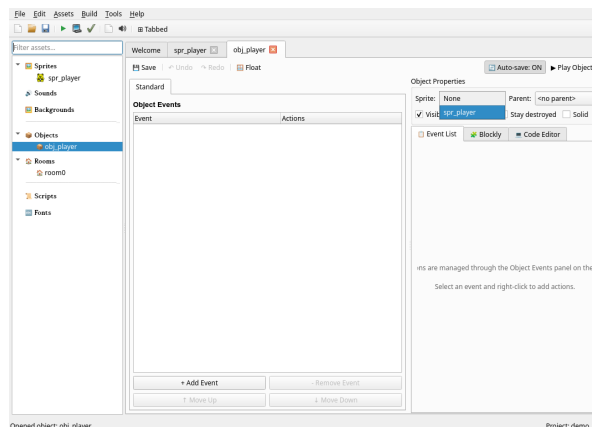
Part 3: Build Your First Project

Now it is time to build something! Follow these eight steps in order, and each one explains what you are doing and why, before telling you what to click.

- ☐ **1. Create a new project.** Every game starts as a project, which is a folder that holds all of its assets. Go to **File > New Project** (or press **Ctrl+N**), give your project a name, and choose where to save it.
- ☐ **2. Create a sprite.** A sprite is the image your character (or any object) will use. Right-click **Sprites**, choose **Create New Sprite**, and give it a name, for example `spr_player`.
- ☐ **3. Edit the sprite.** Double-click the sprite's name, then draw a simple character in the Sprite Editor.



- ☐ **4. Create an object.** A sprite is just a picture carried by an object — an object is what actually behaves in your game. Right-click **Objects**, choose **Create New Object**, and give it a name, for example `obj_player`.
- ☐ **5. Assign a sprite.** Assign a sprite to the object you just created by choosing one from the list shown in the object's properties.



- ☐ **6. Rename the room.** A room is a level or screen in your game. Right-click the room `room0` and rename it `room_game`. This will be your very first level.
- ☐ **7. Place an object.** An object only appears in the game once it has been placed in a room. Double-click your room to open the Room Editor, select `obj_player` from the list, and click inside the room to place it.
- ☐ **8. Test the game.** Press **F5** (or click the green triangle ► in the toolbar, or go to **Build > Test Game**). A window will open showing your room, with the object you placed inside it!

Congratulations! You have just created your first PyGameMaker project. It doesn't do much yet — your object won't move or react to anything — but you will learn how to add behaviors in the next tutorials.

Part 4: A Look Ahead

Now that you know the basics, here is a preview of what is coming next. You do not need to do anything now — you can just read this part to know what to look forward to:

- **Adding movement** — use a Keyboard event and actions like "Move in Direction" to make your object respond to key presses.
- **Collision detection** — make objects react when they touch each other, like stopping at a wall.
- **Visual programming with Blockly** — build behaviors by connecting blocks to each other instead of writing code.
- **Multiple rooms** — create several levels and move between them
- **Exporting your game** — turn your finished project into a game other people can play, for example as a web page or an executable file.

Need help? Ask your teacher, or press **F1** in PyGameMaker for documentation.

Vocabulary

Term	What it means
Asset	Any resource in your project — a sprite, sound, object, or room.
Sprite	An image (or animation) used to draw something on screen.
Object	A game entity — what actually appears and behaves in a room.
Room	A level or screen in your game, where objects are placed.

My Notes
